



Prevention of colistin-induced neurotoxicity: a narrative review of preclinical data

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Received: 13 August 2023 / Accepted: 1 December 2023 / Published online: 13 December 2023
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Abstract

Polymyxin E or colistin is an effective antibiotic against MDR Gram-negative bacteria. Due to unwanted side effects, the use of this antibiotic has been limited for a long time, but in recent years, the widespread of MDR Gram-negative bacteria infections has led to its reintroduction. Neurotoxicity and nephrotoxicity are the significant dose-limiting adverse effects of colistin. Several agents with anti-inflammatory and antioxidant properties have been used for the prevention of colistin-induced neurotoxicity. This study aims to review the preclinical studies in this field to prepare guidance for future human studies. The data was achieved by searching PubMed, Scopus, and Google Scholar databases. All eligible pre-clinical studies performed on neuroprotective agents against colistin-induced neurotoxicity, which were published up to September 2023, were included. Finally, 16 studies (ten in vitro and eight in vivo) are reviewed. Apoptosis (in 13 studies), inflammatory (in four studies), and oxidative stress (in 14 studies) pathways are the most commonly reported pathways involved in colistin-induced neurotoxicity. The assessed compounds include non-herbal (e.g., ascorbic acid, rapamycin, and minocycline) and herbal (e.g., curcumin, rutin, baicalein, salidroside, and ginsenoside) agents. Besides these compounds, some other measures like transplantation of mitochondria and the use of nerve growth factor and mesenchymal stem cells could be motivating subjects for future research. Based on the data from experimental (in vitro and animal) studies, a combination of colistin with neuroprotective agents could prevent or decrease colistin-induced neurotoxicity. However, well-designed randomized clinical trials and human studies are essential for demonstrating efficacy.

Keywords Colistin · Neurotoxicity · Neuroprotective agents · Antioxidants · Polymyxins · Colistimethate sodium

Introduction

Polymyxins, a group of bacteriostatic antibiotics, were discovered from different species of bacterium *Bacillus polymyxa* in 1947. Up to now, more than 15 varieties of polymyxins have been identified, among which the most important ones are polymyxin B and polymyxin E (Mmatli et al. 2020; Nang et al. 2021; Poirel et al. 2017, RL and J 2018).

They act selectively for Gram-negative bacteria because the primary molecular target of these antibacterial agents is the three-domain lipopolysaccharide (O-antigen,

oligosaccharide, and lipid A) which is the main component of the outer membrane of Gram-negative bacteria.

Polymyxin E or colistin has been utilized as a last resort in the treatment of several multidrug resistance Gram-negative bacterial infections such as *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Klebsiella pneumonia* (Ajiboye 2018). Resistance to polymyxins is rare, although there have been increasing reports of polymyxin resistance among carbapenem-resistant gram-negative bacilli (PA 2020, Testing ECOAS 2022).

One of the most important adverse effects of polymyxins is neurotoxicity, which mainly is demonstrated as ataxia, apathy, weakness, confusion, peripheral or orofacial paresthesia, dizziness, visual disturbances, vertigo, seizure and neuromuscular blockade, which can lead to apnea or respiratory failure. Opposite to aminoglycoside-induced neuromuscular blockade, neuromuscular blockade by colistin is a non-competitive blockade, and neostigmine cannot reverse it. In addition, coma, ptosis, dysphagia, psychosis, dysphonia,

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and diplopia can be other neurologic manifestations. The incidence of colistin-related neurotoxicity reported in initial studies was about 7%. However, based on one series, 68% of patients treated with colistin experienced ataxia, paresthesia, or both. However, a systematic review of estimated rates of neurotoxicity in patients treated with polymyxins found no significant difference in the rate of neurotoxicity in this population compared with non-polymyxin-based regimens ($p=0.051$) (Wagenlehner et al. 2021; Bintang et al. 2023). The overall rate of neurotoxicity during polymyxin therapy was 0.030 (95% CI 0.020–0.043). Toxicity was more common in women and happened within 4 days in most patients. Moreover, there is no association between its incidence and patients' age. The CNS toxicity can be reduced by dose reduction (Tsuji et al. 2019). Albeit, these adverse effects are usually benign and reversible (Lim et al. 2010; Ajiboye 2018; MacLaren and Spelman 2012).

Several agents with anti-inflammatory and antioxidant properties such as curcumin and nerve growth factor have been used for the prevention of drug- and toxin-induced neurotoxicity, including colistin. This study aims to review the preclinical studies associated with protective agents in case of colistin neurotoxicity.

Method

The literature was searched in PubMed, Scopus, and Google Scholar databases. Data collection was performed on two occasions, first in July 2022 and then in September 2023, and all relevant articles from commencement to September 2023 were included. MeSH (Medical Subject

Heading) terms and entry terms relating to the topic were used to define the search strategy. The most important terms include “colistin,” “polymyxin,” “CMS,” “colistimathate sodium,” “neurotoxicity,” “neuroprotective agent,” “neural damage,” “neuroprotectant,” and also names of all proposed neuroprotectants against colistin found in the preliminary search. The references of published articles were also screened to find additional relevant studies. The Boolean operators and nesting were as follows: (TS=(Colistin) OR TS=(polymyxin) OR TS=(CMS) OR TS=(colistimathate sodium) OR TS=(polymyxin)) AND (TS=(neurotoxicity) OR TS=(“neuroprotective agent”) OR TS=(“neural damage”) OR TS=(“neuroprotectant”)). We chose the related articles based on their title and abstracts. The inclusion criteria were in vivo and in vitro studies on the prevention of colistin-induced neurotoxicity if it was written in English. We considered no restriction on the availability of the full text if enough data could be accessed from the abstract. Duplicate or unrelated articles were excluded. The search process and initial selection of eligible studies were done by SS and GM.

Results

In this review, 155 articles were found in the search process, while 137 articles were omitted because of duplication and being irrelevant. Finally, 16 articles were reviewed including in vitro ($n=10$, 2 overlapped with in vivo studies) and in vivo ($n=8$) studies (Fig. 1). The included studies are summarized in Table 1.

Fig. 1 Diagram of articles selection process

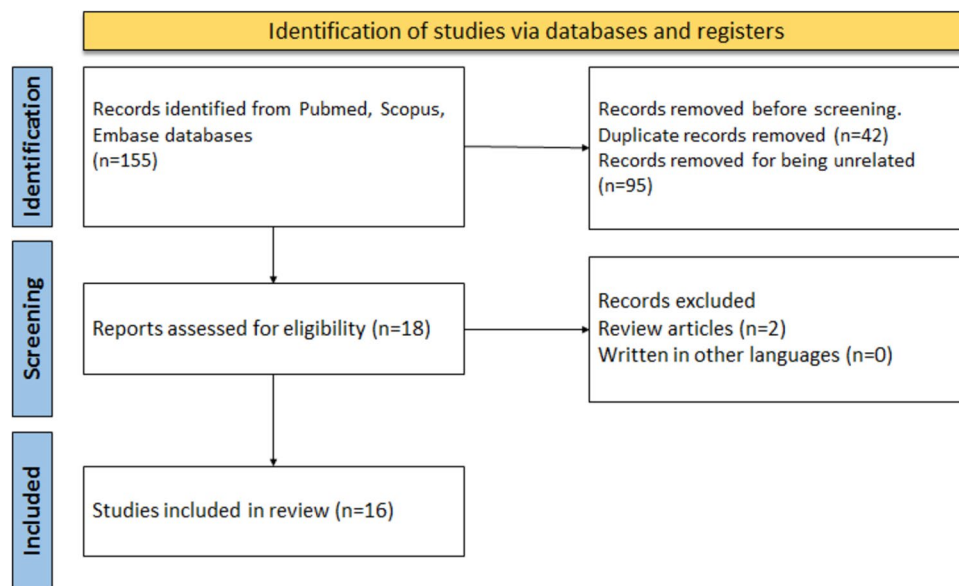


Table 1 Summarized neuroprotective compounds against colistin neurotoxicity

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
Ascorbic acid	PC12 cells Treatment with colistin (125 µg/mL) in presence and absence/presence of ascorbic acid (0.1, 1, 10 µM/mL) for 24 h	In vitro: co-treatment	↓ LDH ⁶ release by ascorbic acid co-treatment		↑ ROS ¹ level & ↓ SOD ² & GSH ³ by COL ⁴ (sig ⁵ . Inhibition of ROS level & ↑ SOD & GSH by ascorbic acid co-treatment in 1, 10 µM/mL concentration)	Sig. ↓ cytochrome-c level By ascorbic acid co-treatment Sig. ↓ caspase-3 & caspase-9 by ascorbic acid co-treatment ↓ DNA fragmentation by ascorbic acid co-treatment	↑ cell viability by ascorbic acid co-treatment (1 and 10 mM/mL) ↓ Cell morphological damages by COL (by ascorbic acid co-treatment in dose-dependent manner)	(Liu et al. 2013)
Baicalein	PC12 cells incubation with CS ⁷ (62.5–500 µg/mL for 24 h) for CS exposure Pre-incubation with baicalein (0, 25, 50, 100 µg/mL) for 24 h and then exposing to 125 µg/mL CS for 24 h	In vitro: pretreatment	↓ LDH in pre-treated cells by baicalein		↓ ROS generation by pretreatment with baicalein	↓ caspase-3 activity ↓ DNA ⁸ fragmentation by pretreatment with baicalein	Improving morphological changes including: ↓ the number of cells Shrinking cells & aggregation into clusters ↓ axons Appearing vacuoles	(Jiang et al. 2013)
Curcumin	Adult male albino rats (N=24 in 4 groups for 6 days) Control: saline solution p.o. ⁹ Curcumin: 200 mg/kg/day p.o. Colistin: 300,000 IU/kg/day i.p. ¹⁰ CUR ¹¹ + COL	In vivo: co-treatment	Sig. ↑ in IL ¹² -6, TNF-α ¹³ & NO ¹⁴ by COL (sig. ↓ by Cur)	Sig. ↓ in Scr ¹⁵ urea & uric acid by CUR	Sig. ↓ in CAT ¹⁶ & GSH levels by COL (sig. ↑ by CUR) Sig. ↑ in MDA ¹⁷ concentration in brain & renal tissue By COL (↓ by CUR)	Mod. ¹⁸ Upregulation of Bcl ¹⁹ -2 expression in the cerebral & renal cortices by CUR	Neuronal shrinking, meningeal congestion, focal gliosis, perivascular lymphocytic cuffing in cerebral cortex by COL disorganizing some Purkinje cells, congestion of the granular layer and meninges, surrounding with perineural vacuolations in cerebellar cortex by COL	(Edrees et al. 2018)

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
Curcumin	Female C57BL/6 mice (<i>n</i> = 10 in each group for 7 days) Control: saline Curcumin: 200 mg/kg/day p.o Colistin: 18 mg/kg/day; i.p COL (18 mg/kg/day) + CUR (50 mg/kg/day) COL + CUR COL + CUR (100 mg/kg/day) COL + CUR (200 mg/kg/day)	In vivo: co-treatment		Sig. ↑ in ATP ²⁰ levels & Na ⁺ -K ⁺ -ATPase activities by CUR-co-administration in the sciatic nerves	Sig. ↑ in MDA level & ↓ in SOD & CAT by COL (sig. ↓ in MDA & ↑ in SOD & CAT by CUR co-administration)	Sig. ↑ in Akt, NGF ²¹ , Nrf2 ²² , HO-1 ²³ & mTOR ²⁴ mRNA expression levels and sig. ↓ in Bax, caspase-3 and caspase-9 mRNA, and activity by CUR co-administration	Sig. improving colistin-induced impaired sensory and motor dysfunctions in a dose-dependent manner through CUR co-administration	(Dai et al. 2020b)
Curcumin	Mouse N2a cells Colistin: 200 μM exposure for 24 h Curcumin: pretreatment at 5, 10, and 20 μM for 2 h prior	In vitro: pretreatment	Sig. ↓ COX-2 ²⁵ , p-IkB-α ²⁶ , NF-κB ²⁷ in dose dependent manner by pretreatment of CUR		Sig. ↓ intracellular ROS and ↑ SOD, CAT & GSH by pretreatment of CUR	↑ Nrf2 & HO-1 by pretreatment of CUR Sig. ↑ Bax, caspase-3 and caspase-9 activity by COL (Sig. ↓ by CUR-pretreatment)	Sig. ↓ colistin-induced cytotoxicity & ↑ the cell viability by CUR	(Dai et al. 2018a)
Ginsenoside Rg1	PC12 cells 5 different groups for 24 h Control Colistin: 125 μg/mL COL + ginsenoside Rg1(5, 10, & 20 μmol/L)	In vitro: co-treatment		↑ MDA & ROS level, Sig. ↓ SOD & GSH activity by COL (sig. reverse by ginsenoside Rg1 co-treatment)	↑ MDA & ROS level, Sig. ↓ SOD & GSH activity by COL (sig. reverse by ginsenoside Rg1 co-treatment)	Sig. ↓ cytochrome-c level By ginsenoside Rg1 co-treatment Sig. ↓ caspase-3 & caspase-9 by ginsenoside Rg1 co-treatment	↑ cell viability by ginsenoside Rg1 co-treatment (10 and 20 μmol/L) Sig. ↓ COL-induced LDH release by ginsenoside Rg1 co-treatment	(Jiang and Li 2014)

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
Mesenchymal stem cells	Adult female albino rats (N=40 in 4 groups) Control MSC: A single dose of 1 × 10 ⁶ /mL through the tail vein Colistin: 36 mg/kg/d for 7 days COL + MSC	In vivo: co-treatment	Sig. ↑ TNF-α, Beclin 1, IL-6, & NGF by COL (restoration by MSC ²⁸ administration)				↑ GFAP ²⁹ by COL and restored by MSC	(Öz Gergin et al. 2023)
Minocycline	Cultured mouse N2a cells and primary cortical neuronal cells Colistin: 200 µM exposure for 24 h Minocycline: pretreatment at 5, 10, and 20 µM for 2 h prior	In vitro: pretreatment			↑ intracellular ROS levels by COL (sig. ↓ intracellular ROS in a dose dependent manner by pretreatment of minocycline) sig. ↑ in SOD & CAT activities by pretreatment of minocycline	Sig. ↓ caspase-3 & caspase-9 activities by pretreatment of minocycline	Sig. ↓ colistin-induced cytotoxicity & ↑ the cell viability by minocycline	(Dai et al. 2017b)
Nerve growth factor	C57BL/6 mice (N = 60 in 6 groups for 7 days) Control NGF: 36 µg/kg/day i.p Colistin: 18 mg/kg/day i.p COL + NGF (9 µg/kg/day) COL + NGF (18 µg/kg/day) COL + NGF (36 µg/kg/day)	In vivo: co-treatment			Sig. ↑ MDA & ↓ SOD & CAT activities by COL (↓ MDA & ↑ SOD & CAT activities by NGF co-treatment)	↑ caspase-3 & caspase-9 by COL (recovery & decrease by NGF co-treatment) Sig. upregulation of HO-1, Bax expression & downregulation of Akt expression by COL (↓ Bax expression & ↑ Akt & HO-1 by NGF co-treatment)	Sig. ↓ thermal pain threshold in a dose-dependent manner by NGF co-treatment ↑ SNCV ³⁰ by NGF co-treatment Thin & loose myelin sheath of the nerve fibers by COL (mark. improving by NGF co-treatment in a dose dependent manner) Mark. ³¹ mitochondrial injury by COL (↓ mitochondrial damage by NGF co-treatment)	(Dai et al. 2020c)

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
Nerve growth factor	PC12 cells in 5 groups for 12 & 24 h Control: DMEM without FBS Colistin: 250 µg/mL COL (250 µg/mL) + NGF (25 ng/mL) COL + NGF (50 ng/mL) COL + NGF (100 ng/mL) Pretreatment with NGF 30 min prior to colistin-treatment	In vitro: pretreatment				↑ LC-3, Beclin 1 & Caspase-3 level by COL (sig. inhibition of these genes by treatment of NGF in a dose dependent manner)	↑ cells viability by co-treatment of NGF both at 12 h and 24 h (sig. by high dose of NGF) IC ₅₀ of colistin: 289.3 µg/mL Mark. ↓ colistin-induced autophagy by NGF at 12 h ↑ apoptotic cell bodies for 12 & 24 h by COL (↓ apoptosis for 12 & 24 h by NGF treatment)	(Tian et al. 2019)
Rapamycin	In vivo: Female C57BL/6 mice (N = 60 in 6 groups for 14 days) Control CS: 18 mg/kg/day i.p (Divided over two 8 hourly doses of 9 mg/kg) Rapamycin: COL + RAP ³² (2.5 mg/kg/day, 2 h prior to COL) In vitro: Mouse N2a cells pre-treatment with rapamycin (100, 200, or 500 nM) for 2 h, following by treatment with COL (200 µM) for 24 h	In vivo: co-treatment In vitro: pretreatment	↑ p-AMPK ³³ , ULK1 ³⁴ & ↓ p-mTOR, & p-CREB ³⁵ by COL treatment (Reversed effects on p-MTOR & p-AMPK & exacerbating the effects of COL on ULK1 by RAP co-treatment)		↓ ROS, ↑ SOD & CAT activities, & GSH Levels in N2a cells by RAP pre-treatment	Sig. upregulation of Bax, Bcl-2, CytC ³⁶ , p-Akt cleaved-caspase-9 and cleaved-caspase-3 proteins, Nrf2, HO-1, LC3II/LC1, p62 & Beclin 1 proteins by COL Mark. ↑ GFP-LC3 dot formation by COL treatment (100 µM) for 24 h in N2a cells (↓ caspase-9 and -3 activities by RAP co-treatment) ↑ Δψm in cerebral cortex	Deeply stained nuclei, neuronal cell damage & mitochondrial injury by COL mark. ↓ colistin-induced mitochondrial damage by RAP slight mitochondrial cristae deletion in COL + RAP ↓ viability of cultured mouse N2a cells by COL (sig. ↑ cellular viability by RAP pre-treatment)	(Dai et al. 2017a)

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
Rutin	male SD ³⁷ rats (8–10 weeks of age) (<i>N</i> = 35 in 5 groups) Control: physiological saline p.o Rutin: 100 mg/Kg/BW p.o Colistin: 15 mg/Kg/BW i.p COL + RUT ³⁸ 50 COL + RUT 100	In vivo: co-treatment	Sig. ↓ TNF- α , nNOS ³⁹ & NF- κ B by Col + RUT		Sig. ↑ in SOD, CAT and GSH in COL + RUT 100 Sig. ↑ MDA by COL ↓ AChE ⁴¹ & BChE ⁴² activities by Rut in dose dependent manner	↑ Nr1f2 in COL + Rut groups Sig. ↓ colistin-increased p53 & caspase3 activity by Rut treatment in a dose dependent manner Sig. ↓ GFAP in Rut groups ↑ BDNF ⁴³ in COL-group Sig. downregulation of BDNF in COL + RUT groups Sig. upregulation of Bcl-2 in COL + RUT groups	Apoptotic neurons & necrotic pyramidal cells in hippocampal area in COL-group Moderate pyknotic changes & apoptotic neurons in hippocampal area of Col + RUT groups ↓ necrotic & apoptotic neurons in hippocampal area ↑ ERK1/2 immunopositivity in hippocampal neurons ↑ CREB reactivity in hippocampal neurons	(Çelik et al. 2020)
Salidroside	RSC96 Schwann cells Colistin induced in various concentration (62.5–1000 µg/mL) for 12, 24, 36, & 48 h Treatment of COL-induced cells by salidroside (0, 3.125, 6.25, 12.5, 25, 50, 100, 200, & 400 µg/mL)	In vitro: pretreatment	↓ LDH release by salidroside		sig. ↓ ROS & ↑ SOD activity by pretreatment of salidroside	↑ levels of p-Akt and Bcl-2 ↓ Bax, release of Cyt-c and activation of caspase-3 by salidroside	Sig. ↓ cell viability by COL in a dose- and time-dependent manner (sig. ↓ cell injury in COL-induced cells by salidroside ↓ mitochondrial damage by inhibiting caspase-3 and activation of Bax	(Lu et al. 2017a)

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
PG ⁴⁴	In vitro: N2a cells COL (3.13, 6.25, 12.5, 25, 50, 100, 200, 400, 800, 1600 µM) and PG (1.953, 3.91, 7.81, 15.62, 31.25, 62.5, 125, 250 µM) for 6, 12, 24, and 48 h (nerve cell damage models were established by 400 µM COL for 24 h)	In vitro: pretreatment In vivo: pretreatment			↓ ROS level by PG Sig. ↓ MDA levels by PG Preventing the decrease in CAT, GSH, and SOD levels induced by COL	Sig. ↓ Cyt-c, ↓ Bax, caspase-3 and -9 ↑ Bcl-2 ↑ NRF2 and pNRF2 protein levels, Upregulation of HO-1 protein levels by PG	Sig. ↓ COL-induced neuronal apoptosis levels by PG Reversing the deviations in mitochondrial morphology and membrane potential by PG Suppressing autophagy levels in N2a cells by PG ↓ COL-induced oxidative stress and mitochondrial damage via activating the NRF2/HO-1 pathway by PG ↑ cell viability by PG	(Xiong et al. 2023)
	IC ₅₀ of COL: 254 µM In vivo: Male Kunming mice (8 weeks of age) pretreatment with vehicle or PG (4 mg/kg and 16 mg/kg) by tail vein injection 2 h before i.v. injection of COL, then, 2 µL colistin (8000 U/kg/d and 16,000 U/kg/d) was injected into the lateral ventricle for five days at a rate of 0.3 µL/min through the sleeve using a microinjection pump							

Table 1 (continued)

Compound	Study design	Type of study	Inflammatory markers	Biochemical markers	Antioxidant markers	Apoptotic markers	Histological effect	Ref
SDCS ⁴⁵	Male mice C57BL/6 (6–8 weeks of age) (<i>N</i> =36 in 4 groups for 7 days) Control: saline solution i.p Colistin:15 mg/kg/day i.p F1(CO:SDCS 1:1); 15 mg/kg/day COL equivalent i.p F2 (COL: SDCS 1:2) i.p	In vivo: Co-treatment			Sig. ↓ SOD in F1 and F2 group		Sig. ↓ thermal pain threshold in SDCS incorporated groups ↓ fatty acyl exposure and lowering the charge of COL by SDCS	(Bintang et al. 2023)
Healthy Mitochondria	rat renal proximal tubular cells (RPTCs) The control group: DMSO (% 0.05) The COL group: 120 µg/ml of colistin The COL + healthy mitochondria group: 120 µg/ml of colistin for 2 h/ 100 µg/ml of healthy mitochondria The healthy mitochondria group:100 µg/ml of healthy mitochondria	In vitro: co-treatment			Sig. ↓ ROS & ↑ GSH by co-treatment with mitochondria	Sig. ↓ caspase-3	Sig. ↓ colistin-induced cytotoxicity & ↑ the cell viability (based on LDH assay) Sig. restored the loss of mitochondrial membrane integrity & lysosomal damage	(Arjmand et al. 2023)

(1) Reactive oxygen species, (2) Superoxide dismutase, (3) Glutathione, (4) Colistin, (5) Significantly, (6) Lactate dehydrogenase, (7) colistin sulfate, (8) deoxyribonucleic acid, (9) orally, (10) intraperitoneal, (11) curcumin, (12) interleukin, (13) tumor necrosis factor alpha, (14) nitric oxide, (15) serum creatinine, (16) catalase, (17) malondialdehyde, (18) moderately, (19) B-cell lymphoma, (20) adenosine triphosphate, (21) nerve growth factor, (22) nuclear factor erythroid 2-related factor 2, (23) heme oxygenase 1, (24) mammalian/mechanistic target of rapamycin, (25) cyclooxygenase-2, (26) p-inhibitor of NF-κB, (27) Nuclear factor kappa B, (28) mesenchymal stem cell, (29) glial fibrillary acidic protein, (30) sensory nerve conduction velocity, (31) markedly, (32) rapamycin, (33) AMP-activated protein kinase, (34) unc-51-like kinase 1, (35) cAMP response element-binding protein, (36) cytochrome C, (37) Sprague Dawley, (38) rutin, (39) neuronal nitric oxide synthase, (40) glutathione peroxidase, (41) acetylcholinesterase, (42) butyrylcholinesterase, (43) brain-derived neurotrophic factor, (44) piceatannol-3'-O-β-D glucopyranoside, (45) sodium deoxycholate sulfate

Mechanisms of colistin-induced neurotoxicity

Colistin-induced neurotoxicity and neuromuscular blockage occur via two principal mechanisms: presynaptic activity and biphasic depolarization. The former prevents acetylcholine release into the synaptic gap, whereas the latter leads to calcium loss from neurons, mitochondrial dysfunction, reactive oxygen species accumulation, and nerve damage (Camargo et al. 2021).

Recent studies indicate that colistin-induced neurotoxicity is mainly related to mitochondrial dysfunction and oxidative stress (Dai et al. 2017b, Dai et al. 2013a, Liu et al. 2013, Dai et al. 2018b, Dai et al. 2016, Dai et al. 2013b), although the data shows that multiple inflammatory and pro/anti-apoptotic signaling pathways including PI3K, Akt, p53, mitogen-activated protein kinase, adenosine monophosphate (AMP)-activated protein kinase, mammalian target of rapamycin C1 (mTORC1), NRF2/HO-1, Bax, and Bcl-2 are involved in colistin-induced neuronal cell death as well (Dai et al. 2017b, Dai et al. 2016, Dai et al. 2013b, Dai et al. 2018a, Dai et al. 2017a). The consequences of colistin neurotoxicity on different markers are shown in Fig. 2.

Neuroprotective compounds

Ascorbic acid

Ascorbic acid is a potent water-soluble antioxidant and a cofactor for many enzymes (Lee et al. 2020). About 90% of daily intake in the general population comes from fruit and vegetables. The content varies among species, but mango, kiwi, citrus fruits, and vegetables such as peppers, tomatoes, and broccoli are rich sources of vitamin C (Lykkesfeldt et al. 2014).

This compound acts as a scavenger of free radicals such as aqueous peroxy, superoxide anions, and hydroxyl radicals. Ascorbic acid can also neutralize reactive oxygen species (ROS) in the extracellular space by contributing an electron to reduce free radicals before reacting with biological molecules (Carr and Frei 1999; Evans and Halliwell 2001). Previous investigations have revealed that its strong antioxidant ability can effectively prevent colistin-induced nephrotoxicity in both pre-clinical and human studies (Jafari and Elyasi 2021); however, clinical data on ascorbic acid ability to prevent colistin-induced neurotoxicity is lacking.

Yang Liu and his team examined the protective effect of ascorbic acid against colistin-induced neurotoxicity in PC12 cells. The PC12 cells were exposed to colistin (125 µg/mL) with or without the ascorbic acid for 24 h. Intracellular ROS accumulation resulting from colistin administration was considerably decreased by ascorbic acid.

Based on this study, ascorbic acid at the concentrations of 1 and 10 µM/mL could effectively enhance the activity

of superoxide dismutase (SOD) and increase the glutathione (GSH) level, which led to the reduction of intracellular ROS in colistin treatment PC12 cells.

In addition, ascorbic acid reversed the increase in DNA fragmentation, a biochemical hallmark of apoptosis, caused by colistin. Moreover, ascorbic acid significantly decreased the expression of caspase-3 and -9 and inhibited the release of mitochondrial cytochrome c in colistin-treated PC12 cells (Liu et al. 2013).

So it seems that ascorbic acid can be presented as a promising neuroprotectant for colistin-induced neurotoxicity, mediated by oxidative stress. Nevertheless, further in vivo experiments are essential for better judgment, followed by well-designed human studies in case of positive results of animal studies.

Baicalein

Baicalein is an important type of flavonoid component extracted from the dried root of *Scutellaria baicalensis* Georgi (Jiang et al. 2013). Many in vivo and in vitro studies have shown baicalein possess several pharmacological effects including anticancer, antithrombotic, anti-inflammatory, antioxidant, antidiabetic, hepatoprotective, cardioprotective, nephroprotective, and neuroprotective properties (Dinda et al. 2017; Liang et al. 2017; Sowndhararajan et al. 2018; Jafari and Elyasi 2021).

Several studies have exhibited that baicalein effectively prevents many neurodegenerative diseases such as Parkinson's disease, Alzheimer's disease, and cerebral ischemia through decreasing oxidative stress and apoptosis, inhibiting excitotoxicity and inflammatory cytokines production, and promoting neurogenesis and cell differentiation (Liang et al. 2017; Cao et al. 2010; Sowndhararajan et al. 2018). Many phenolic hydroxyl groups exist in the chemical structural formula of baicalin, which is strongly associated with free radical (including hydroxyl, superoxide, and alkyl) scavenging activity (Jiang et al. 2013).

A study by Jiang et al. on the neuroprotective effects of baicalein for the neurotoxicity of colistin in PC12 cells indicated that pretreatment with baicalein considerably reduced colistin-related cell death. An increased dose of colistin sulfate can significantly alter the cell morphology such as decreasing the number of cells, shrinking cells and aggregation into clusters, decreasing axons, and appearing vacuoles. In addition, according to this study, preincubation with baicalein at doses of 50 and 100 µg/ml for 24 h significantly reduced the lactate dehydrogenase release levels in comparison to cells treated with colistin sulfate alone. As it mentioned before, colistin sulfate can dose-dependently increase caspase-3, an apoptosis regulatory protein, activity. Treatment with baicalein for 24 h before PC12 cell's exposure to colistin sulfate markedly suppressed caspase-3

activity besides the generation of reactive oxygen species and lactate dehydrogenase lev in a dose-dependent manner which can lead to prevention of apoptosis (Jiang et al. 2013).

In sum, these results highlight the prospect of baicalein to alleviate colistin-induced neurotoxicity. Jiang et al. proposed that a concentration of 50 µg/mL baicalin can be added to colistin sulfate injections to protect against colistin-induced neurotoxicity, and simultaneously, it can be used to control bacterial infections and increase the clinical curative effect of colistin.

Curcumin

Curcumin is an anti-inflammatory compound extracted from the rhizome of *Curcuma longa* and the gum resin of *Boswellia serrata*. Several studies demonstrated variety of its pharmacological and biological activities such as anti-inflammatory, antioxidant, and immune-regulatory activities, as well as preventive effects against cancers (Liu et al. 2018; Carroll et al. 2011), diabetes (Chuengsamarn et al. 2012), and cardiovascular (Hernández et al. 2016; Yu et al. 2016) and nervous system (Dikmen et al. 2017), kidney (de Almeida Alvarenga et al. 2018), and liver diseases (Nabavi et al. 2014). It has also shown a synergistically inhibitory effect on the growth of bacteria in combination with polymyxin (Kaur et al. 2018; Betts et al. 2016). On the other hand, it can alleviate colistin-induced nephrotoxicity and neurotoxicity by inhibiting mitochondrial dysfunction, oxidative stress, apoptosis, and inflammation (Ammon et al. 1993; Dai et al. 2020a; Xu et al. 2018).

In a study done by Dai and his team in 2016, a potential role for curcumin has been suggested to treat polymyxin-induced neurotoxicity through the modulation of NF-κB signaling and its potent anti-oxidative and anti-apoptotic effects. Using a murine model in 2020 demonstrated that co-administration of oral curcumin remarkably improved colistin-induced motor dysfunctions and impaired sensory in a dose-dependent manner. Curcumin not only can markedly upregulate SOD and catalase (CAT) activities, ATP levels, Na⁺/K⁺-ATPase activity, levels of AKT, heme oxygenase 1 (HO-1) mRNA, mTOR, and Nrf2, but also can decrease the expression of p-IκB-α and COX-2 and downregulate Bax, caspase-3 and -9 mRNA, and lipid peroxide compared to the colistin alone group in sciatic nerve tissue. Colistin induces inflammatory response via IL-1β and NF-κB pathway signaling and subsequently induction of the COX-2 enzyme. NF-κB also regulates the expression of more than 500 genes related to inflammation, tumorigenesis, and cellular proliferation. So, curcumin as NF-κB inhibitor could be effective in the prevention of colistin-induced adverse reactions. It involves the downregulation of NF-κB signaling and the mediators of inflammation (e.g., COX-2) by inhibiting the activity of IKK kinase. So, this investigation supports the neuroprotective effects of curcumin during colistin treatment (Dai et al. 2018a; Dai et al. 2020b).

Ginsenoside Rg1

Ginseng, the root of *Panax* species, has been traditionally used as an herbal drug for thousands of years. Ginsenosides,

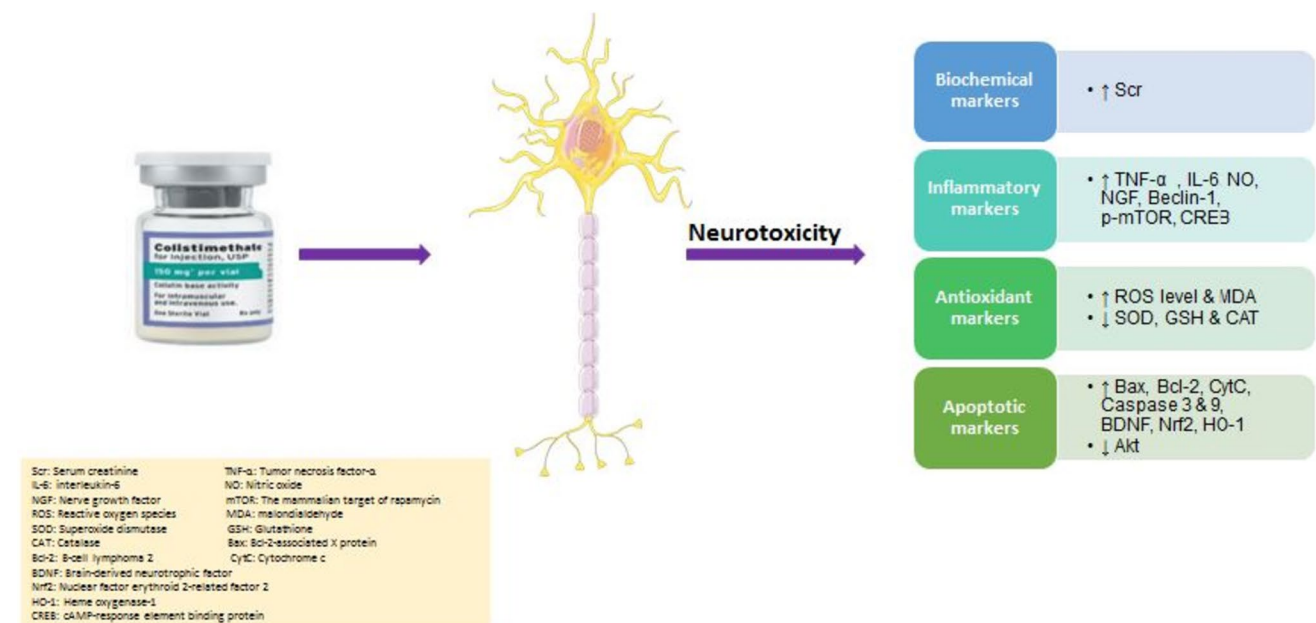


Fig. 2 Probable mechanisms of colistin neurotoxicity

a group of steroidal saponins, are recognized to be the major bioactive ingredients of ginseng (Cheng et al. 2019; Song et al. 2017; Gao et al. 2020). They have various properties such as neuroprotection and cardioprotection, and also work as an anti-diabetes, anti-tumor, and anti-inflammatory agent.

Ginsenosides are classified into dammarane-type, also as protopanaxadiol (PPD) and protopanaxatriol (PPT), ocotillol-type, oleanane-type, and polysaccharides (Gao et al. 2020). The medicinal ingredients of ginseng include Rb1, Rb2, Rc, Rd, Re, Rf, Rg1, Rg3, Rh2, F1, compound K (CK), PPD, and PPT (Jin et al. 2019).

Compared to other ginsenosides, ginsenoside Rg1 has not only multi-target effects but also fewer adverse consequences.

A systematic preclinical review conducted by Song et al. considered that ginsenoside Rg1 could enhance the neurobehavioral abnormality and show potential neuroprotective effects on the Parkinson's disease model via several mechanisms, including antioxidant stress, anti-neuroinflammation, and anti-apoptosis (Song et al. 2017).

In addition, Rg1 could improve axonal atrophy, memory disorders, and synaptic loss in an Alzheimer's disease model induced by amyloid β peptide (Tohda et al. 2004).

Based on a study by Gou et al., co-treatment with ginsenoside Rg1 prevented apoptotic cell death in PC12 cells via inhibition of oxidative stress and mitochondrial pathway in a dose-dependent manner. Accordingly, colistin administration caused a considerable elevation of oxidative stress through the increase of ROS and malondialdehyde (MDA) levels and the decrease in GSH and SOD activities. Co-incubation of PC12 cells with ginsenoside Rg1 can enter mitochondria and stabilize the mitochondrial redox balance, which leads to attenuating the foregoing changes. In addition, ginsenoside Rg1 could reduce the expression of caspase-3 and caspase-9 and release of mitochondrial cytochrome c in PC12 cells treated with colistin, causing apoptosis inhibition (Jiang and Li 2014).

In summary, ginsenoside Rg1 can protect against colistin-induced neurotoxicity; however, well-designed randomized clinical trials will help to better judgment.

Mesenchymal stem cell

Mesenchymal stem cells (MSCs) are multipotent and self-renewing cells that can differentiate into several types of cells such as osteocytes, chondrocytes, and adipocytes. MSC mainly isolates from bone marrow, but other tissues like the placenta, amniotic fluid, adipose tissue, and umbilical cord blood can be other sources of MSC (Kim and Cho 2013).

Several studies have exhibited that exogenous administration of stem cells considerably reduces brain damage in intraventricular hemorrhage (IVH), neonatal stroke, and hypoxic-ischemic encephalopathy (HIE) models.

MSCs can secrete anti-inflammatory cytokines and different growth factors, including brain-derived neurotrophic factor (BDNF), vascular endothelial growth factor (VEGF), NGF, and basic fibroblast growth factor, which are responsible for mediating the proliferation and differentiation of progenitor cells. In addition, MSCs can inhibit cell apoptosis as well as decrease inflammation and oxidative stress.

According to a research, MSC treatment upregulates the expression of genes related to cell proliferation and neurogenesis (NGF) while downregulates genes associated with inflammation such as interleukin (IL)1- β in brain-damaged areas (Nair et al. 2021).

Research by Oz Gergin et al. in 2022 about the effects of MSC on colistin-induced neurotoxicity approved that colistin treatment increases glial fibrillary acidic protein (GFAP), NGF, and Beclin 1, and cytokines' immune reactivity intensity including IL-6 and tumor necrosis factor- α (TNF- α) and co-administration with MSC can partly restore these markers. This study demonstrated that MSCs ameliorate apoptosis, inflammation, and autophagy due to colistin and could be a promising agent for preventing colistin neurotoxicity (Öz Gergin et al. 2023).

Minocycline

Minocycline, a semisynthetic antibiotic derivate of tetracycline, has been used for many years as an antibacterial agent. Evidence shows that in addition to antimicrobial properties, this agent possesses anti-apoptotic, anti-inflammatory, antioxidant, and neuroprotective effects. Minocycline can pass easily through the blood–brain barrier. This feature makes it an interesting drug candidate for treating many neurological diseases (Wang et al. 2021).

Studies demonstrated that minocycline, as an inhibitor of caspase-3, can stabilize mitochondrial membranes by decreasing their permeability. In addition, it can reduce Bid protein release and neuron apoptosis and inhibit microglial activation (Wang et al. 2021).

Previous experiments reported that colistin administration not only markedly increases intracellular ROS levels but also decreases the activity of the antioxidant enzymes CAT and SOD (Dai et al. 2017b).

One of the characteristic properties of minocycline is antioxidant activity which is associated with its ability to chelate mitochondrial iron, which catalyzes toxic hydroxyl radical formation (Hu et al. 2016a). Based on Dai et al. research minocycline significantly attenuates colistin-induced neurotoxicity in mouse N2a and primary cortical neuronal cells. They proposed that colistin-induced apoptosis in neuronal N2a cells is activated via both the death receptor (extrinsic) and the mitochondrial (intrinsic) pathways. Caspase-3 is a key apoptotic mediator that can be activated by both of these pathways. Caspase-9 is also an important mediator in

the mitochondrial apoptosis pathway. Minocycline pretreatment significantly downregulated the activities of caspase-3 and -9 in colistin-treated N2a cells. Moreover, minocycline could inhibit colistin-induced ROS generation and improve the total antioxidant capacity of N2a cells via upregulation of the activities of CAT and SOD and inhibition of colistin-induced ROS generation (Dai et al. 2017b).

In conclusion, minocycline therapy may depict a promising novel approach for preventing neurotoxicity in patients receiving colistin.

Nerve growth factor

Nerve growth factor (NGF) belongs to the neurotrophins family (Manni et al. 2013) which are essential in the development and maintenance of the central and peripheral nervous system.

NGF has been used as a neuroprotectant in the treatment of several neurodegenerative diseases including toxin and drug-mediated neurotoxicity (Tan et al. 2019; Salinas et al. 2003; Liu et al. 2016).

Many studies showed that NGF could promote the differentiation and survival of different types of neurons via PI3K/Akt and MAPK/ERK pathways. NGF through two kinds of receptors, namely tyrosine kinase receptor A (TrkA) and p75 neurotrophin receptor, can activate foregoing downstream signaling pathways. In addition, NGF can decrease neural damage caused by chemical hypoxia through the reduction of extreme amounts of calcium ions, oxidative stress, and toxic amino acids (Zhang et al. 2011; Oliveira et al. 2013; Hu et al. 2016b; Nguyen et al. 2009; Kirschner et al. 1996).

Moreover, it has been considered that NGF can activate Nrf2 translocation into the nucleus and restrain oxidative damage caused by B-amyloid in SKNSH neuroblastoma cells (Su et al. 2019). The binding of Nrf2 can promote the transcription activity of antioxidant response elements and lead to the expression of antioxidant mediators including SOD, CAT, and HO-1 (Magesh et al. 2012). According to a recent study by Dai et al., NGF co-administration upregulated colistin-induced HO-1 gene expression and SOD and CAT activities in sciatic nerve tissues of mice (Dai et al. 2020c).

Although available data about the inhibition of autophagy by NGF is limited, many researches demonstrate that NGF can induce protective autophagy to downregulate apoptosis.

Based on a study by Tian et al., nerve growth factor can protect PC12 cells against colistin-induced neurotoxicity. Additionally, this factor reduced apoptosis and suppressed autophagy caused by colistin and partly recovered mitochondrial membrane potential, and improved the viability of PC12 cells in a dose-dependent manner by TrkA receptor through PI3K/Akt (Tian et al. 2019).

Furthermore, research indicated that colistin treatment could lead to its accumulation in the cerebellum, brain, and sciatic nerve tissues, which induces harmful ultrastructural changes and dysfunction in the mitochondria of these tissues. Bax and caspase-9 are two significant biomarkers of the mitochondrial apoptotic pathway in neuronal cell death (Dai et al. 2019; Dai et al. 2020c). NGF can inhibit apoptotic cell death downstream and upstream of mitochondrial outer membrane permeabilization (MOMP) and post-caspase activation through the removal of active caspase-3 in a lysosome-dependent manner (Mnich et al. 2014).

In summary, the mentioned studies revealed that NGF co-administration ameliorates colistin-induced neurotoxicity via not only inhibiting apoptosis and oxidative stress but also decreasing autophagy. The underlying mechanism may be associated with upregulating the Akt and Nrf2/HO-1 pathways.

NGF could be a promising adjuvant therapy in preventing colistin-induced neurotoxicity although further in vivo studies are needed.

Rapamycin

Rapamycin, a lipophilic macrocyclic lactone, produced by the bacterium *Streptomyces hygroscopicus*, has potent immunosuppressive properties. Rapamycin has been approved by the FDA for treating lymphangioleiomyomatosis by preventing the rejection of organ transplantation and coating coronary stents. It also has shown different pharmacological activities, such as inhibiting the serine/threonine kinase, which is termed mTOR, as well as reducing neuronal apoptotic cell death using the Akt pathway in the Parkinson's disease model; therefore, it was demonstrated that this substance could be used in the means of enervating oxidative stress (Dai et al. 2017a).

It has been observed that by disrupting cytokine signaling, rapamycin can promote lymphocyte growth and differentiation. Moreover, the G1/S transition of the proliferation cycle in IL-2-stimulated T cells can affect the phosphorylation/activation of the p70 S6 kinase (p70s6k), and by inhibiting it, rapamycin will reduce the transition of specific mRNA and thereby decrease the protein synthesis. On the other hand, via the same pathway, it can inhibit the activity of the cyclin-dependent kinase cdk2-cyclin E complex (Dumont and Su 1995).

Activated p70S6K and mTORC1, which is a mechanistic target for rapamycin and is part of a regulatory-associated complex responsible for regulating cell growth in response to nutrients, can significantly elevate the hippocampus and cerebral cortex in animal models being studied in Alzheimer's disease and other neurological diseases and therefore being suggested as a neuroprotective compound. In research done by Dai and his team, the effect of rapamycin

in reducing oxidative stress caused by colistin *in vivo* and *in vitro* has been observed. In the mouse models treated with colistin (18 mg/kg/day divided into two doses of 9 mg/kg, cumulative dose 252 mg/kg), neuronal mitochondria damage in the cerebral cortex and an increase in the caspase-9 and -3 activation were demonstrated; the examination was followed by rapamycin (2.5 mg/kg/day, cumulative dose 35 mg/kg) pre-treatment reducing the neurotoxic effect. In the *in vitro* research using mouse neuroblastoma-2a (N2a) cell culture model, rapamycin has shown a neuroprotective effect by decreasing intracellular ROS production and upregulating anti-ROS activities of enzyme catalase and superoxide dismutase. Amelioration of mitochondrial dysfunction and rescue from apoptosis via the activation of autophagy and mTOR/Akt/CREB signaling pathway are other proposed mechanisms of action. Rapamycin can directly affect mitochondrial function by regulating the transcription of genes involved in mitochondrial redox homeostasis, such as TORC1, YY1, and PGC-1 α . It decreased the expression of Bax and the activation of caspase-9 and -3, thereby attenuating apoptosis. Moreover, it attenuated colistin-induced activation of p53 (Dai et al. 2017a; Dai et al. 2018b).

So, it can be concluded that rapamycin can effectively reduce colistin-induced neurotoxicity in neuronal cells through inhibition of mitochondrial dysfunction, apoptosis, and ROS production.

Rutin

Flavonoids are a family of phenolic compounds with several protective effects including antioxidant, antineoplastic, anti-inflammatory, anti-allergic, antibacterial, antithrombotic, and neuroprotective properties (Panche et al. 2016). Rutin (quercetin-3-O-rutinoside) is one of the most important members of this family. It exists in many nutrient sources, such as fruits, vegetables, grains, green tea, wine, and most citrus species. Rutin can protect against lung and liver damage by reducing inflammation and oxidative stress via modulation of the MAPK pathway (Song et al. 2018). Previous studies have reported that rutin has various neuropharmacological applications such as antidepressant, anticonvulsant, and neuroprotective activities on the central nervous system. However, its mechanism of action is still under investigation. Some results indicate that the chemical structure of rutin can directly scavenge ROS. Besides, it restrains xanthine oxidase, which is related to the generation of ROS. Also, rutin can increase the production of oxidative stress-related markers, including GSH, SOD, and CAT. The mechanisms mentioned can be responsible for rutin antioxidant activity. A study conducted by Çelik et al. on the effect of rutin on the neurotoxicity of colistin in rats demonstrated that colistin-related decrease in SOD, CAT, GPx enzyme, and level of GSH was improved by rutin co-treatment. In addition, the

levels of MDA that increased in the colistin group were considerably reduced by rutin (Çelik et al. 2020). Moreover, rutin noticeably reduced the levels of TNF- α , NF- κ B, and nNOS in colistin-induced neurotoxicity (Çelik et al. 2020).

Mitochondria have an important role in the induction of apoptosis (Cheng et al. 2012). Anti- and pro-apoptotic Bcl-2 family proteins, such as Bcl2 and Bax, can regulate the mitochondrial pathway (Dai et al. 2016). Colistin increases Bax/Bcl2 ratios and cytochrome C release, leading to the activation of caspase-3 and -9 that trigger apoptosis (Dai et al. 2016; Dai et al. 2017a). Çelik's study indicates that rutin co-treatment can decrease caspase-3, Bax and cytochrome c expressions.

In conclusion, rutin supplementation can alleviate colistin-induced neurotoxicity. The probable mechanisms may be associated with anti-inflammatory, antioxidant, and anti-apoptosis effects of rutin in brain tissue.

Salidroside

Salidroside is a predominant ingredient isolated from the plant *Rhodiola rosea* L., known as Hongjingtian in Chinese, and has been used in traditional Chinese medicine to maintain body health. This precious herbaceous plant, known as a medicinal plant for a long time, is distributed at high altitudes in the Arctic and mountainous regions of Europe and Asia. The main compounds extracted from this medicinal herb are salidroside and tyrosyl. The concentrations of these two important active components in *Rhodiola* L. are standard indices for evaluating the quality of *Rhodiola* L. In recent years, many biological activities, such as adaptogenic, anti-hypoxia, anti-fatigue, enhancement of learning and memory, and anticancer and antioxidant effects, have been demonstrated (Mao et al. 2007; Kang et al. 1998; Spasov et al. 2000; Zheng and Wang 2001).

Further studies suggest that salidroside, as one of the significant compounds, also have different pharmacological effects such as anti-aging (Gen-Xiang et al. 2010), memory improvement (Palmeri et al. 2016), anti-fatigue (Li et al. 2008), antioxidant (Zhang et al. 2007), and anti-depressant effects (Van Diermen et al. 2009).

In addition, Salidroside has been found to have neuroprotective properties, reducing glutamate-induced apoptosis in rat hippocampal neurons and alleviating beta-amyloid injury in SH-SY5Y human neuroblastoma cells (Lu et al. 2017a, Chen et al. 2008; Zhang et al. 2010).

Moreover, salidroside can control the release of neurotransmitters that are important for mood regulation and cognitive function, including dopamine, norepinephrine, and serotonin.

Recent investigations have shown that salidroside protects cardiac H9c2 cells from hydrogen peroxide injury and pc12 cells from methyl-4-phenylpyridinium (MPP $^{+}$)-induced

apoptosis through activation of PI3k/Akt pathway (Zhu et al. 2011; Zhang et al. 2012). The PI3k/Akt pathway is a crucial survival factor in the peripheral and central nervous system, effectively regulating mitochondrial pathway-induced cell apoptosis. The Akt pathway, activated by PI3k in response to cytokines, insulin, growth factor, and cell stress, enhances cell tolerance during free radical exposure, oxidative stress, and DNA damage (Tasaki et al. 2010, Lu et al. 2017b).

A study by Lu et al. about the protective effect of salidroside against colistin-induced neurotoxicity in RSC96 Schwann cells indicates that pretreatment with salidroside reverses the effects of colistin on Akt phosphorylation inhibition, promotes Bcl-2 expression, reduces Cyt c release and caspase-3 activation, and inhibits Bax activity. Furthermore, salidroside may mitigate the oxidative stress caused by colistin by stabilizing mitochondrial redox balance by directly scavenging ROS along with decreasing electron leakage and subsequent ROS formation (Dai et al. 2018b, Lu et al. 2017a).

Piceatannol-3'-O- β -D-glucopyranoside (PG)

Piceatannol-3'-O-D-glucopyranoside (PG), a glucoside derivative of resveratrol, is the principal bioactive substance extracted from *Rheum undulatum* L. According to preclinical research, antioxidant, anti-apoptosis, and anti-inflammatory qualities of PG are helpful to protect against brain ischemia. Five resveratrol derivatives were tested in vitro for their capacity to directly capture methylglyoxal and scavenge free radicals; PG demonstrated the strongest activity (Liu et al. 2020). The BBB is easily crossed by PG, opening up a potentially effective route for neuroprotection.

Xiong et al. have accomplished in vitro and in vivo investigations on the possible impact of PG on colistin-induced neurotoxicity. Colistin causes severe damage to N2a cells, time and dose-dependently. After being exposed to 125 μ M PG for 24 h, cell viability was decreased by about 22%. These results suggest that PG, at various doses, is safe for N2a cells. Cell viability was 52.99% and 30.41%, respectively, after 24 h of exposure to 200 and 400 μ M colistin which was prevented by treatment with PG (62.5 μ M) ($P < 0.05$ and 0.01, respectively).

Moreover, the apoptotic rate of N2a cells was raised in a dose-dependent manner by two dosages of colistin (200 and 400 μ M). Apoptosis rates fell significantly from 13.9 to 7.62% ($P < 0.0001$) and 3.9% ($P < 0.0001$), respectively, after pretreatment of N2a cells with PG at 5 and 50 μ M for 6 h before 400 μ M colistin exposure.

The intracellular ROS levels were dramatically increased in N2a cells after being exposed to 400 μ M colistin for 24 h ($P < 0.0001$) that reduced after treatment with PG (5, 50 μ M) ($P < 0.0005$ and < 0.0001 , respectively). Additionally, colistin raised MDA levels but decreased GSH, CAT, and SOD

activity ($P < 0.0001$). The pretreatment of N2a cells with 5 and 50 μ M PG for 6 h before colistin significantly inhibited the rise in MDA levels ($P < 0.0001$) and avoided the decline in CAT, GSH, and SOD levels. These findings show that oxidative stress contributes to colistin-induced neuronal cell damage and that PG possesses strong antioxidant properties.

Colistin significantly lowered the potential of the mitochondrial membrane ($P < 0.0001$), while treatment of N2a cells with PG (5, 50 μ M) dose-dependently enhanced the ratio of red to green fluorescence ($P < 0.0001$). Also, using the fluorescent indicator MitoTracker Red CMXRos demonstrated that the usual morphology of the wounded mitochondria was recovered after treatment with PG.

The primary indicators of mitochondrial damage and apoptosis are an elevated Bax/Bcl-2 ratio and intracellular Cyt c release. The expressions of pro-apoptotic proteins, such as Bax and Cyt c ($P < 0.0001$), and caspase-3/-9 were noticeably increased ($P < 0.0005$), while those of the anti-apoptotic protein (Bcl-2) were dramatically repressed, after colistin (400 μ M) treatment of N2a cells for 24 h. Pretreatment with PG, on the other hand, resulted in an increase in anti-apoptotic protein levels and a suppression of pro-apoptotic protein levels, notably cytochrome c levels ($P < 0.0001$).

Kelch-like ECH-associated protein 1 (Keap-1) controls NRF2 expression by promoting NRF2 proteasomal degradation through contact with ubiquitin ligases. Meanwhile, a crucial step in controlling antioxidant activity is the translocation of phosphorylated NRF2 (pNRF2) into the nucleus. Colistin blocked the NRF2/HO-1 pathway, which decreased N2a ability to act as an antioxidant. In cells and the nucleus, pretreatment with 50 μ M PG raised NRF2 protein levels ($P < 0.0001$) as well as pNRF2 protein levels ($P < 0.001$). Moreover, HO-1 protein expression levels increased ($P < 0.001$), whereas Keap-1 protein expression levels remained unaltered.

The activation of autophagy, which controls NRF2 signaling, can stop cells from dying from colistin-induced apoptosis. Autophagy levels were elevated in colistin-treated N2a cells, supporting earlier findings (Dai et al. 2017a; Zhang et al. 2016). In N2a cells, treatment with PG increased p62 levels ($P < 0.01$, $P < 0.0001$) and lowered the ratio of LC3B II/I ($P < 0.05$, $P < 0.01$). In N2a cells treated with PG, pULK1/ULK1 levels rose. These results suggest that PG inhibited the colistin-induced activation of autophagy. Also, it was demonstrated using Nrf2 knockdown N2a cell models that PG significantly reduced the nerve damage brought on by colistin by activating the NRF2 pathway.

Colistin-induced neurotoxicity after intraventricular injection was assessed in a mice model. Neurons in the CA1 region of the hippocampus in the high-dose colistin group (16,000 U/kg/day) that had previously received PG had triangular pyrosis. Besides, the Nissl bodies in the

degenerating cells were significantly diminished or even absent. Some neurons had also undergone severe denature, creating vacuoles. Following colistin therapy, the levels of oxidative stress in brain tissue considerably rose, as seen by decreased GSH, SOD, and CAT levels but elevated MDA levels. Treatment with PG (4 and 6 mg/kg 2 h before colistin injection) prevented the oxidative damage that colistin causes. Additionally, caspase-3 and cytochrome c expression levels that were stimulated by the colistin drug were substantially decreased by PG (Xiong et al. 2023).

Sodium deoxycholate sulfate (SDCS)

Preclinical studies demonstrated that sodium deoxycholate sulfate (SDCS) effectively diminished the nephrotoxicity of amphotericin B and polymyxin B. Drugs incorporated into SDCS made stable micelles and showed slower release and consequently lower exposure rate and fewer nephro- and neurotoxicity. Besides, SDCS did not hinder the antibacterial activity of polymyxins. Bintang et al. examined the neurotoxic and nephrotoxic effects of colistin compounded with internally produced SDCS in a mouse model. Mice randomly were assigned to four groups: saline solution as the control, colistin (15 mg/kg/day), colistin/SDCS 1:1 (F1), and colistin/SDCS 1:2 (F2). Comparative to the control group, the colistin-exposed mice developed sensory neuropathies. Compared to the colistin group, the SDCS-included formulations significantly lowered the thermal pain threshold after 7 days of treatment. When compared to the colistin group (10.42 s), which was significantly higher ($P < 0.01$) than the control group (7.32 s), the thermal pain threshold values for the F1- and F2-injected mice reduced to 8.03 s and 8.01 s, respectively.

Moreover, serum SOD levels significantly dropped in the F1 (0.72 ± 0.03 U/mL) and F2 (0.71 ± 0.06 U/mL) groups ($P < 0.05$) in comparison to the control group (0.52 ± 0.05 U/mL). Similar patterns were seen in the serum CAT levels of the colistin-SDCS-treated animals ($P < 0.05$). Regarding histopathological examination, in the colistin-treated animals, infiltration of monocytes with loss of hepatocyte architecture around the blood vessels was observed. The histological changes were less severe with a few monocyte invasions in the F1- and F2-treated animals. Finally, they suggested that SDCS reduces fatty acyl exposure and lowers the charge of colistin, which might reduce the damage of colistin in the sciatic nerve tissue (Bintang et al. 2023).

Transplantation of mitochondria

Several studies show the association of mitochondrial dysfunction with colistin-induced adverse reactions, including neurotoxicity. It seems that inhibiting mitochondrial dysfunction or replacing the damaged mitochondria via

increasing the number of mitochondria (fission and fusion) and mitochondrial transplantation can be an appropriate approach to the prevention of its toxicities. Transplantation of mitochondria is a novel approach to the treatment of mitochondrial dysfunction induced by drugs, chemicals, and inherited or acquired metabolic disorders. In Arjmand et al.'s study, they isolated rat renal proximal tubular cells (RPTCs) from the kidney and exposed them to a toxic concentration of colistin with/without rat healthy isolated mitochondria for 4 h. The results showed that administration of isolated mitochondria could reduce mitochondrial dysfunction, activity of LDH, production of ROS, ATP, and GSH depletion, loss of MMP, lysosomal damages, and cell death (Arjmand et al. 2023). However, this study is not published yet and more studies in this field are necessary for better judgment.

Conclusion

This review article provides an overview of various agents utilized to prevent or attenuate colistin neurotoxicity. Based on assessed studies, stress oxidative besides activating inflammatory and apoptotic pathways is the main mechanism of colistin neurotoxicity. The activation of the p53, death receptor pathways, and mitochondrial dysfunction usually are followed by apoptosis.

The compounds including ascorbic acid, baicalein, mesenchymal stem cell, nerve growth factor, curcumin, ginsenoside Rg1, minocycline, rapamycin, rutin, PG, and SDCS demonstrated their protective effects through different mechanisms, but apoptosis (in 13 studies), inflammatory (in four studies), and oxidative stress (in 14 studies) pathways are the most common reported pathways. We examined preclinical studies (nine in vitro and eight in vivo studies) in this article. Co-treatment (10 studies) and pretreatment (eight studies) with agents with antioxidant properties mentioned above can reverse these markers in a dose-dependent manner. In addition, these compounds mostly have antiapoptotic effects, which can affect hallmarks of apoptosis such as Bax, caspase-3, caspase-9, Nrf2, and HO-1. Moreover, colistin-induced neurotoxicity involves the inflammatory response via IL-1 β and NF- κ B pathway signaling with subsequent induction of the inflammatory enzyme COX-2. NF- κ B subunits translocate into the nucleus and bind to the promoters of target genes encoding pro-inflammatory cytokines (IL-1, IL-2, IL-6, TNF- α) and pro-inflammatory enzymes including COX-2 and inducible nitric oxide synthase after its activation. So, for prevention of this adverse reaction, the researches should be focused on compounds with well-known effects on these pathways via these abovementioned mediators.

It should be also mentioned that based on our previous review (Jafari and Elyasi 2021), some of these

compounds including vitamin C, curcumin, and baicalein were effective measures for the prevention of colistin-induced nephrotoxicity. Vitamin C and curcumin are mostly famous for their antioxidant effect and baicalein besides its antioxidant effect possesses anti-inflammatory and anti-apoptotic properties. Confirming their efficacy in human studies could propose them as good options for the prevention of both nephro- and neurotoxicity of colistin. To the best of our knowledge, the only available RCT in this field is performed on vitamin C nephroprotective effect, which did not show promising findings (Sirijatuphat et al. 2015). However, a prospective cohort observational study on patients with sepsis shock or severe sepsis showed that the adjuvant use of vitamin C with a dose of 2 g BD IV may be effective in the prevention of colistin-induced nephrotoxicity, at least in this specific population of patients (Dalfino et al. 2015). Furthermore, vitamin C has pharmacokinetic interaction with colistin and increases its half-life by reducing its clearance and consequently its efficacy and/or its toxicity which should be considered. So, the significant efficacy of compounds in in vivo studies should be confirmed in well-designed RCTs. Animal studies are performed under controlled conditions on completely healthy animals. So, their findings could not be generalized to humans. Well-designed RCTs on these compounds for the prevention and management of colistin neurotoxicity are necessary for better judgment. Considering the specialty of the authors which are clinical pharmacists, at first, we searched for human studies on this topic. However, as no clinical trials or another type of human studies in this field were available, we provide this review focusing on pre-clinical data to encourage the researchers to design and implement human studies. Moreover, most of these compounds just have oral formulations and colistin usually is used as the last resort in critically ill patients which lots of whom are NPO. So, preparation of IV formulation or oral formulation which could easily be given by oral gavage may be helpful. As the high dose of IV vitamin C is used commonly in critically ill patients as an antioxidant, it could be a good choice for future human studies.

Rapamycin which showed promising efficacy in the prevention of neurotoxicity in preclinical studies, actually does not seem an appropriate candidate as it poses considerable adverse reactions and limited indications. However, minocycline which is an antibiotic with good efficacy against *Acinetobacter* could be considered an interesting option besides colistin, for both double coverage of *Acinetobacter* and reducing colistin-induced neurotoxicity. Even retrospective studies assessing the incidence of neurotoxicity in patients who received colistin beside minocycline in comparison with colistin alone could be suggested for the first step.

Finally, besides chemical and herbal compounds, some other measures like transplantation of mitochondria and the use of nerve growth factor and mesenchymal stem cells could be motivating subjects for future research.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1007/s00210-023-02884-w>.

Author contributions Setareh Soroudi and Ghazal Mousavi searched the databases and wrote the manuscript. Fatemeh Jafari reviewed and edited the manuscript. Sepideh Elyasi defined the manuscript's subject and finalized the manuscript. The authors confirm that no paper mill and artificial intelligence was used.

Data availability Not applicable.

Declarations

Ethical approval Not applicable.

Competing interests The authors declare no competing interests.

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